

Avacnes2.5

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```
datos <-read.csv("test.csv")
```

```
##      Id SalePrice economico medio caro
## 1 1461  169277.1         1      0     0
## 2 1462  187758.4         0      1     0
## 3 1463  183583.7         1      0     0
## 4 1464  179317.5         1      0     0
## 5 1465  150730.1         1      0     0
## 6 1466  177151.0         1      0     0
```

##	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour
## 1	1	60	RL	65	8450	Pave	<NA>	Reg	Lvl
## 2	2	20	RL	80	9600	Pave	<NA>	Reg	Lvl
## 3	3	60	RL	68	11250	Pave	<NA>	IR1	Lvl
## 4	4	70	RL	60	9550	Pave	<NA>	IR1	Lvl
## 5	5	60	RL	84	14260	Pave	<NA>	IR1	Lvl
## 6	6	50	RL	85	14115	Pave	<NA>	IR1	Lvl
##	Utilities	LotConfig	LandSlope	Neighborhood	Condition1	Condition2	BldgType		
## 1	AllPub	Inside	Gtl	CollgCr	Norm	Norm	1Fam		
## 2	AllPub	FR2	Gtl	Veenker	Feedr	Norm	1Fam		
## 3	AllPub	Inside	Gtl	CollgCr	Norm	Norm	1Fam		
## 4	AllPub	Corner	Gtl	Crawfor	Norm	Norm	1Fam		
## 5	AllPub	FR2	Gtl	NoRidge	Norm	Norm	1Fam		
## 6	AllPub	Inside	Gtl	Mitchel	Norm	Norm	1Fam		
##	HouseStyle	OverallQual	OverallCond	YearBuilt	YearRemodAdd	RoofStyle	RoofMatl		
## 1	2Story	7	5	2003	2003	Gable	CompShg		
## 2	1Story	6	8	1976	1976	Gable	CompShg		
## 3	2Story	7	5	2001	2002	Gable	CompShg		
## 4	2Story	7	5	1915	1970	Gable	CompShg		
## 5	2Story	8	5	2000	2000	Gable	CompShg		
## 6	1.5Fin	5	5	1993	1995	Gable	CompShg		
##	Exterior1st	Exterior2nd	MasVnrType	MasVnrArea	ExterQual	ExterCond	Foundation		
## 1	VinylSd	VinylSd	BrkFace	196	Gd	TA	PConc		
## 2	MetalSd	MetalSd	None	0	TA	TA	CBlock		
## 3	VinylSd	VinylSd	BrkFace	162	Gd	TA	PConc		
## 4	Wd Sdng	Wd Shng	None	0	TA	TA	BrkTil		
## 5	VinylSd	VinylSd	BrkFace	350	Gd	TA	PConc		
## 6	VinylSd	VinylSd	None	0	TA	TA	Wood		
##	BsmtQual	BsmtCond	BsmtExposure	BsmtFinType1	BsmtFinSF1	BsmtFinType2			
## 1	Gd	TA	No	GLQ	706	Unf			
## 2	Gd	TA	Gd	ALQ	978	Unf			
## 3	Gd	TA	Mn	GLQ	486	Unf			
## 4	TA	Gd	No	ALQ	216	Unf			
## 5	Gd	TA	Av	GLQ	655	Unf			
## 6	Gd	TA	No	GLQ	732	Unf			
##	BsmtFinSF2	BsmtUnfSF	TotalBsmtSF	Heating	HeatingQC	CentralAir	Electrical		
## 1	0	150	856	GasA	Ex	Y	SBrkr		
## 2	0	284	1262	GasA	Ex	Y	SBrkr		
## 3	0	434	920	GasA	Ex	Y	SBrkr		
## 4	0	540	756	GasA	Gd	Y	SBrkr		
## 5	0	490	1145	GasA	Ex	Y	SBrkr		
## 6	0	64	796	GasA	Ex	Y	SBrkr		
##	X1stFlrSF	X2ndFlrSF	LowQualFinSF	GrLivArea	BsmtFullBath	BsmtHalfBath	FullBath		
## 1	856	854	0	1710	1	0	2		
## 2	1262	0	0	1262	0	1	2		
## 3	920	866	0	1786	1	0	2		
## 4	961	756	0	1717	1	0	1		
## 5	1145	1053	0	2198	1	0	2		
## 6	796	566	0	1362	1	0	1		
##	HalfBath	BedroomAbvGr	KitchenAbvGr	KitchenQual	TotRmsAbvGrd	Functional			
## 1	1	3	1	Gd	8	Typ			
## 2	0	3	1	TA	6	Typ			

```

## 3      1      3      1      Gd      6      Typ
## 4      0      3      1      Gd      7      Typ
## 5      1      4      1      Gd      9      Typ
## 6      1      1      1      TA      5      Typ
##  Fireplaces FireplaceQu GarageType GarageYrBlt GarageFinish GarageCars
## 1      0      <NA> Attchd      2003      RFn      2
## 2      1      TA      Attchd      1976      RFn      2
## 3      1      TA      Attchd      2001      RFn      2
## 4      1      Gd      Detchd      1998      Unf      3
## 5      1      TA      Attchd      2000      RFn      3
## 6      0      <NA> Attchd      1993      Unf      2
##  GarageArea GarageQual GarageCond PavedDrive WoodDeckSF OpenPorchSF
## 1      548      TA      TA      Y      0      61
## 2      460      TA      TA      Y      298      0
## 3      608      TA      TA      Y      0      42
## 4      642      TA      TA      Y      0      35
## 5      836      TA      TA      Y      192      84
## 6      480      TA      TA      Y      40      30
##  EnclosedPorch X3SsnPorch ScreenPorch PoolArea PoolQC Fence MiscFeature
## 1      0      0      0      0      <NA> <NA> <NA>
## 2      0      0      0      0      <NA> <NA> <NA>
## 3      0      0      0      0      <NA> <NA> <NA>
## 4      272      0      0      0      <NA> <NA> <NA>
## 5      0      0      0      0      <NA> <NA> <NA>
## 6      0      320      0      0      <NA> MnPrv      Shed
##  MiscVal MoSold YrSold SaleType SaleCondition SalePrice economico medio caro
## 1      0      2      2008      WD      Normal      208500      1      0      0
## 2      0      5      2007      WD      Normal      181500      1      0      0
## 3      0      9      2008      WD      Normal      223500      1      0      0
## 4      0      2      2006      WD      Abnorml      140000      1      0      0
## 5      0      12      2008      WD      Normal      250000      1      0      0
## 6      700      10      2009      WD      Normal      143000      1      0      0

```

Se observa que utilizamos para la variable objetivo tres categorías, economico, caro y medio donde primero calculamos el valor de la casa máxima y el mínimo valor de la casa donde luego lo restamos el valor máximo con el mínimo y luego dividirlo entre tres para así diferenciar que casa debería de pertenecer en esas tres variables.

Solo se elimina la variable ID, Exterior2ndCBlock, GrLivArea, TotalBsmtSF porque no es importante en estos análisis y evitar el análisis de más variables.

```
summary(modelo_todas_cv)
```

```

##
## Call:
## NULL
##
## Coefficients: (3 not defined because of singularities)
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    2.657e+01  1.654e+07      0      1
## MSSubClass      6.661e-15  1.301e+03      0      1
## MSZoningFV     -1.533e-13  1.889e+05      0      1
## MSZoningRH      4.699e-13  1.872e+05      0      1
## MSZoningRL      3.488e-13  1.608e+05      0      1
## MSZoningRM      4.195e-13  1.505e+05      0      1
## LotFrontage    -1.545e-14  6.903e+02      0      1
## LotArea        -1.213e-16  1.749e+00      0      1
## StreetPave     -3.577e-12  1.904e+05      0      1
## AlleyPave       1.715e-12  7.469e+04      0      1
## LotShapeIR2    -4.465e-13  6.640e+04      0      1
## LotShapeIR3     1.419e-12  1.398e+05      0      1
## LotShapeReg    -6.592e-13  2.516e+04      0      1
## LandContourHLS -2.003e-12  8.051e+04      0      1
## LandContourLow  1.793e-12  1.001e+05      0      1
## LandContourLvl  2.327e-14  5.794e+04      0      1
## UtilitiesNoSeWa 8.389e-12  4.145e+05      0      1
## LotConfigCulDSac -1.878e-12  5.206e+04      0      1
## LotConfigFR2   -2.475e-13  6.332e+04      0      1
## LotConfigFR3   -2.153e-12  1.976e+05      0      1
## LotConfigInside -2.748e-13  2.814e+04      0      1
## LandSlopeMod    5.262e-13  6.254e+04      0      1
## LandSlopeSev    7.730e-12  1.803e+05      0      1
## NeighborhoodBlueste 8.778e-13  3.032e+05      0      1
## NeighborhoodBrDale -2.786e-15  1.725e+05      0      1
## NeighborhoodBrkSide 1.314e-12  1.489e+05      0      1
## NeighborhoodClearCr 2.268e-12  1.446e+05      0      1
## NeighborhoodCollgCr 2.118e-12  1.141e+05      0      1
## NeighborhoodCrawfor 2.427e-12  1.346e+05      0      1
## NeighborhoodEdwards 3.516e-12  1.261e+05      0      1
## NeighborhoodGilbert 1.773e-12  1.207e+05      0      1
## NeighborhoodIDOTRR 5.281e-12  1.688e+05      0      1
## NeighborhoodMeadowV 1.164e-12  1.763e+05      0      1
## NeighborhoodMitchel 1.662e-12  1.286e+05      0      1
## NeighborhoodNAMES 1.807e-12  1.232e+05      0      1
## NeighborhoodNoRidge 5.046e-12  1.339e+05      0      1
## NeighborhoodNPkVill -5.475e-13  2.213e+05      0      1
## NeighborhoodNridgHt 1.322e-12  1.193e+05      0      1
## NeighborhoodNWAMES 2.114e-12  1.262e+05      0      1
## NeighborhoodOldTown 5.554e-12  1.510e+05      0      1
## NeighborhoodSawyer 2.579e-12  1.278e+05      0      1
## NeighborhoodSawyerW 1.988e-12  1.224e+05      0      1
## NeighborhoodSomerst 1.683e-12  1.418e+05      0      1
## NeighborhoodStoneBr -2.015e-12  1.319e+05      0      1
## NeighborhoodSWISU 4.673e-12  1.525e+05      0      1
## NeighborhoodTimber 1.144e-12  1.275e+05      0      1

```

## NeighborhoodVeenker	3.504e-13	1.653e+05	0	1
## Condition1Feedr	4.815e-12	7.859e+04	0	1
## Condition1Norm	4.107e-12	6.596e+04	0	1
## Condition1PosA	5.164e-12	1.571e+05	0	1
## Condition1PosN	4.473e-12	1.169e+05	0	1
## Condition1RR Ae	4.870e-12	1.427e+05	0	1
## Condition1RR An	3.474e-12	1.090e+05	0	1
## Condition1RR Ne	5.202e-12	2.748e+05	0	1
## Condition1RR Nn	-5.933e-13	2.028e+05	0	1
## Condition2Feedr	2.546e-12	3.681e+05	0	1
## Condition2Norm	-4.530e-12	3.183e+05	0	1
## Condition2PosA	7.785e-12	5.831e+05	0	1
## Condition2PosN	6.791e-12	4.484e+05	0	1
## Condition2RR Ae	1.410e-11	8.951e+05	0	1
## Condition2RR An	-4.819e-12	4.955e+05	0	1
## Condition2RR Nn	-3.474e-12	4.250e+05	0	1
## BldgType2fmCon	-2.640e-12	1.965e+05	0	1
## BldgTypeDuplex	-1.713e-12	1.157e+05	0	1
## BldgTypeTwnhs	-4.017e-13	1.580e+05	0	1
## BldgTypeTwnhsE	3.541e-13	1.422e+05	0	1
## HouseStyle1.5Unf	-5.766e-13	1.243e+05	0	1
## HouseStyle1Story	-1.336e-12	6.860e+04	0	1
## HouseStyle2.5Fin	1.080e-11	1.942e+05	0	1
## HouseStyle2.5Unf	-2.510e-12	1.453e+05	0	1
## HouseStyle2Story	-1.723e-12	5.502e+04	0	1
## HouseStyleSFoyer	-8.396e-13	9.788e+04	0	1
## HouseStyleSLvl	-9.990e-13	8.699e+04	0	1
## OverallQual	1.036e-13	1.631e+04	0	1
## OverallCond	5.208e-13	1.399e+04	0	1
## YearBuilt	-4.551e-15	1.208e+03	0	1
## YearRemodAdd	-3.681e-14	8.758e+02	0	1
## RoofStyleGable	4.064e-12	2.901e+05	0	1
## RoofStyleGambrel	4.350e-12	3.173e+05	0	1
## RoofStyleHip	3.940e-12	2.913e+05	0	1
## RoofStyleMansard	3.202e-12	3.369e+05	0	1
## RoofStyleShed	-4.901e-12	5.456e+05	0	1
## RoofMatlCompShg	-3.391e-11	6.426e+05	0	1
## RoofMatlMembran	-3.073e-11	8.608e+05	0	1
## RoofMatlMetal	-3.811e-11	8.403e+05	0	1
## RoofMatlRoll	-2.966e-11	7.506e+05	0	1
## `RoofMatlTar&Grv`	-2.907e-11	7.013e+05	0	1
## RoofMatlWdShake	-3.427e-11	6.880e+05	0	1
## RoofMatlWdShngl	-3.471e-11	6.686e+05	0	1
## Exterior1stAsphShn	-5.949e-12	5.153e+05	0	1
## Exterior1stBrkComm	1.691e-12	4.351e+05	0	1
## Exterior1stBrkFace	1.219e-12	1.994e+05	0	1
## Exterior1stCBlock	8.826e-12	4.274e+05	0	1
## Exterior1stCemntBd	-3.590e-11	3.040e+05	0	1
## Exterior1stHdBoard	-5.367e-13	2.023e+05	0	1
## Exterior1stImStucc	-1.644e-11	4.436e+05	0	1
## Exterior1stMetalSd	4.740e-13	2.288e+05	0	1
## Exterior1stPlywood	-7.452e-13	1.998e+05	0	1

## Exterior1stStone	6.152e-12	3.767e+05	0	1
## Exterior1stStucco	-4.716e-12	2.215e+05	0	1
## Exterior1stVinylSd	2.108e-12	2.081e+05	0	1
## `Exterior1stWd Sdng`	4.409e-12	1.931e+05	0	1
## Exterior1stWdShing	1.537e-12	2.092e+05	0	1
## Exterior2ndAsphShn	4.137e-12	3.485e+05	0	1
## `Exterior2ndBrk Cmn`	6.902e-13	3.151e+05	0	1
## Exterior2ndBrkFace	-1.247e-12	2.060e+05	0	1
## Exterior2ndCBlock	NA	NA	NA	NA
## Exterior2ndCmentBd	3.728e-11	3.001e+05	0	1
## Exterior2ndHdBoard	1.716e-12	1.945e+05	0	1
## Exterior2ndImStucc	2.061e-11	2.252e+05	0	1
## Exterior2ndMetalSd	1.114e-12	2.229e+05	0	1
## Exterior2ndOther	4.558e-12	4.270e+05	0	1
## Exterior2ndPlywood	1.553e-12	1.889e+05	0	1
## Exterior2ndStone	-1.339e-12	2.668e+05	0	1
## Exterior2ndStucco	-5.230e-12	2.143e+05	0	1
## Exterior2ndVinylSd	-6.318e-13	2.010e+05	0	1
## `Exterior2ndWd Sdng`	-2.578e-12	1.870e+05	0	1
## `Exterior2ndWd Shng`	-1.936e-12	1.955e+05	0	1
## MasVnrTypeBrkFace	-2.249e-12	1.076e+05	0	1
## MasVnrTypeNone	-2.203e-12	1.086e+05	0	1
## MasVnrTypeStone	-2.595e-12	1.138e+05	0	1
## MasVnrArea	-3.952e-15	9.123e+01	0	1
## ExterQualFa	8.387e-12	1.741e+05	0	1
## ExterQualGd	7.456e-12	7.594e+04	0	1
## ExterQualTA	9.024e-12	8.380e+04	0	1
## ExterCondFa	5.801e-12	2.824e+05	0	1
## ExterCondGd	1.060e-12	2.699e+05	0	1
## ExterCondPo	9.093e-12	4.948e+05	0	1
## ExterCondTA	2.250e-12	2.693e+05	0	1
## FoundationCBlock	-3.620e-13	4.991e+04	0	1
## FoundationPConc	-2.265e-12	5.381e+04	0	1
## FoundationSlab	8.230e-13	1.233e+05	0	1
## FoundationStone	2.471e-12	1.769e+05	0	1
## FoundationWood	-2.084e-12	2.325e+05	0	1
## BsmtQualFa	-2.740e-12	9.951e+04	0	1
## BsmtQualGd	-3.467e-12	5.389e+04	0	1
## BsmtQualTA	-3.393e-12	6.598e+04	0	1
## BsmtCondGd	-1.943e-12	8.297e+04	0	1
## BsmtCondPo	-1.524e-11	4.703e+05	0	1
## BsmtCondTA	-1.665e-12	6.669e+04	0	1
## BsmtExposureGd	-2.608e-12	4.779e+04	0	1
## BsmtExposureMn	3.545e-14	4.756e+04	0	1
## BsmtExposureNo	-1.427e-13	3.429e+04	0	1
## BsmtFinType1BLQ	5.532e-13	4.386e+04	0	1
## BsmtFinType1GLQ	4.038e-13	3.971e+04	0	1
## BsmtFinType1LwQ	9.449e-13	5.885e+04	0	1
## BsmtFinType1Rec	1.548e-12	4.710e+04	0	1
## BsmtFinType1Unf	4.131e-13	4.550e+04	0	1
## BsmtFinSF1	6.287e-17	7.473e+01	0	1
## BsmtFinType2BLQ	3.962e-13	1.187e+05	0	1

## BsmtFinType2GLQ	-1.447e-12	1.474e+05	0	1
## BsmtFinType2LwQ	-4.191e-12	1.162e+05	0	1
## BsmtFinType2Rec	-4.805e-13	1.118e+05	0	1
## BsmtFinType2Unf	1.102e-13	1.183e+05	0	1
## BsmtFinSF2	2.161e-15	1.343e+02	0	1
## BsmtUnfSF	-1.396e-16	6.293e+01	0	1
## TotalBsmtSF	NA	NA	NA	NA
## HeatingGasA	2.104e-12	3.867e+05	0	1
## HeatingGasW	4.472e-12	4.006e+05	0	1
## HeatingGrav	5.751e-13	4.259e+05	0	1
## HeatingOthW	3.451e-12	4.850e+05	0	1
## HeatingWall	8.756e-12	4.500e+05	0	1
## HeatingQCFA	-1.317e-12	7.416e+04	0	1
## HeatingQCgd	-6.874e-13	3.253e+04	0	1
## HeatingQCPo	-1.435e-12	4.180e+05	0	1
## HeatingQCTA	1.930e-13	3.258e+04	0	1
## CentralAirY	-1.391e-12	6.081e+04	0	1
## ElectricalFuseF	1.177e-13	9.041e+04	0	1
## ElectricalFuseP	-1.714e-12	2.899e+05	0	1
## ElectricalMix	1.564e-12	7.000e+05	0	1
## ElectricalSBrkr	-2.232e-13	4.626e+04	0	1
## X1stFlrSF	-2.889e-16	8.657e+01	0	1
## X2ndFlrSF	-1.529e-15	9.389e+01	0	1
## LowQualFinSF	-4.523e-14	2.954e+02	0	1
## GrLivArea	NA	NA	NA	NA
## BsmtFullBath	-3.463e-13	3.110e+04	0	1
## BsmtHalfBath	-2.140e-12	4.758e+04	0	1
## FullBath	-1.709e-13	3.469e+04	0	1
## HalfBath	9.954e-13	3.297e+04	0	1
## BedroomAbvGr	3.263e-13	2.152e+04	0	1
## KitchenAbvGr	1.719e-12	8.914e+04	0	1
## KitchenQualFa	-2.590e-12	9.784e+04	0	1
## KitchenQualGd	-1.837e-12	5.585e+04	0	1
## KitchenQualTA	-1.554e-12	6.257e+04	0	1
## TotRmsAbvGrd	-3.200e-13	1.504e+04	0	1
## FunctionalMaj2	6.646e-12	2.265e+05	0	1
## FunctionalMin1	-1.694e-12	1.354e+05	0	1
## FunctionalMin2	-2.800e-12	1.357e+05	0	1
## FunctionalMod	-1.147e-11	1.661e+05	0	1
## FunctionalSev	2.264e-14	4.649e+05	0	1
## FunctionalTyp	-2.455e-12	1.176e+05	0	1
## Fireplaces	3.104e-13	2.401e+04	0	1
## FireplaceQuFa	9.691e-12	1.083e+05	0	1
## FireplaceQuGd	9.266e-12	8.392e+04	0	1
## FireplaceQuPo	9.801e-12	1.233e+05	0	1
## FireplaceQuTA	9.822e-12	8.695e+04	0	1
## GarageTypeAttchd	-2.049e-12	1.735e+05	0	1
## GarageTypeBasment	-1.629e-12	2.004e+05	0	1
## GarageTypeBuiltIn	-3.053e-12	1.806e+05	0	1
## GarageTypeCarPort	-3.931e-12	2.305e+05	0	1
## GarageTypeDetchd	-2.283e-12	1.729e+05	0	1
## GarageYrBlt	5.361e-14	8.923e+02	0	1

## GarageFinishRFn	2.004e-13	3.098e+04	0	1
## GarageFinishUnf	3.763e-14	3.773e+04	0	1
## GarageCars	1.250e-12	3.453e+04	0	1
## GarageArea	-6.820e-15	1.224e+02	0	1
## GarageQualFa	-9.740e-12	4.832e+05	0	1
## GarageQualGd	-1.227e-11	4.941e+05	0	1
## GarageQualPo	-1.076e-11	6.125e+05	0	1
## GarageQualTA	-1.024e-11	4.782e+05	0	1
## GarageCondFa	1.361e-11	5.528e+05	0	1
## GarageCondGd	1.559e-11	5.731e+05	0	1
## GarageCondPo	1.534e-11	5.924e+05	0	1
## GarageCondTA	1.328e-11	5.482e+05	0	1
## PavedDriveP	2.144e-12	8.597e+04	0	1
## PavedDriveY	2.028e-13	5.379e+04	0	1
## WoodDeckSF	-1.217e-15	9.257e+01	0	1
## OpenPorchSF	-5.785e-15	1.806e+02	0	1
## EnclosedPorch	-2.526e-14	1.961e+02	0	1
## X3SsnPorch	-2.427e-16	3.520e+02	0	1
## ScreenPorch	-1.085e-14	1.973e+02	0	1
## PoolArea	9.296e-15	4.636e+02	0	1
## PoolQCFa	-2.021e-10	5.004e+05	0	1
## PoolQCGd	-2.018e-10	4.034e+05	0	1
## FenceGdWo	3.809e-12	7.735e+04	0	1
## FenceMnPrv	3.733e-12	5.704e+04	0	1
## FenceMnWw	4.546e-12	1.292e+05	0	1
## MiscFeatureOthr	2.505e-11	1.006e+06	0	1
## MiscFeatureShed	1.930e-11	1.038e+06	0	1
## MiscFeatureTenC	1.812e-11	1.070e+06	0	1
## MiscVal	1.252e-15	6.304e+01	0	1
## MoSold	8.678e-14	3.858e+03	0	1
## YrSold	-1.270e-14	8.101e+03	0	1
## SaleTypeCon	-1.459e-12	2.768e+05	0	1
## SaleTypeConLD	1.039e-12	1.526e+05	0	1
## SaleTypeConLI	-2.007e-12	1.813e+05	0	1
## SaleTypeConLw	-7.088e-13	1.908e+05	0	1
## SaleTypeCWD	-1.570e-12	2.027e+05	0	1
## SaleTypeNew	3.616e-12	2.427e+05	0	1
## SaleTypeOth	3.002e-13	2.255e+05	0	1
## SaleTypeWD	-9.568e-13	6.570e+04	0	1
## SaleConditionAdjLand	-1.266e-12	2.276e+05	0	1
## SaleConditionAlloca	-7.232e-13	1.364e+05	0	1
## SaleConditionFamily	5.142e-13	9.558e+04	0	1
## SaleConditionNormal	2.484e-13	4.570e+04	0	1
## SaleConditionPartial	-4.573e-12	2.333e+05	0	1
## SalePrice	5.242e-17	5.647e-01	0	1
## economico	-5.313e+01	2.113e+05	0	1
## medio	-5.313e+01	1.853e+05	0	1
##				
## (Dispersion parameter for binomial family taken to be 1)				
##				
## Null deviance: 9.9264e+01 on 1459 degrees of freedom				
## Residual deviance: 8.4703e-09 on 1214 degrees of freedom				


```
## AIC: 492
##
## Number of Fisher Scoring iterations: 25
```

El modelo de regresión logística con validación cruzada mostró que ninguna de las variables es significativa, ya que los valores de z son 0 y los valores de $\Pr(>|z|)$ son 1, lo que indica que las variables no están contribuyendo de manera relevante. Esto podría ser causado por la multicolinealidad entre las variables, debido a que muchas de ellas están altamente correlacionadas, lo que afecta la estimación de los coeficientes. Otra posible causa es el exceso de variables en relación con el número de observaciones, lo que podría llevar a un sobreajuste lo cual es más probable con los modelos realizadas con anterioridad.